

POLICY BRIEF

DATATHON 2026

Team Name: 5Gen data

Track: 2

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WHAT THIS NOTE IS ABOUT

The Karnataka education system needs to figure out the best way to help students to learn basic math in government schools. This brief suggests whether districts should get general support for foundational learning or specific help for certain competencies, depending on how their students are performing. Instead of using the same approach, we should put resources where each district struggles most with math.

THE EVIDENCE

We looked at the datasets and grouped questions into the nine math competencies based on the performance data. We calculated scores for each district and saw the weakest competency and how big the gap was between their best and worst competencies.

Our analysis shows that Measurement (49.7%), Place Value (50.3%), and Division (51.1%) are the weakest competencies across the districts. These three competencies are a huge disadvantage in 28 out of 30 districts.

We also saw a clear pattern in how students performed in arithmetic operations:

1. Addition: 63.9%
2. Subtraction: 57.3%
3. Multiplication: 54.5%
4. Division: 51.1%

This steady drop indicates that students find it harder as math concepts get more complicated.

When we looked at districts, we found three main groups:

1. 11 districts where Measurement has the weakest competency.
2. 9 districts where Division has the weakest competency.
3. 8 districts where Place Value has the weakest competency.

WHY IT MATTERS

As districts have different needs, we can use their specific challenges for guiding suggestions. Some need general help, while others need work on particular competencies. Focusing on specific competencies makes teacher training, catch up programs and learning materials more effective.

RECOMMENDATIONS

1. Focus teacher training on Measurement in Kalaburgi, Chitradurga, Raichur and Koppal.
2. Help Kodagu and Mandya students with Division as that's their main issue.
3. Give extra support in Raichur and Yadgir with things like teacher mentoring, math programs and learning tools.
4. In future tests, track progress on specific competencies not just overall scores.

FEASABILITY AND SCALABILITY

This plan uses information we already have and the training systems that are already in place, so it won't cost extra for data collection. Districts with similar problems can be grouped and given the same support model, making it easy for affordability and wider implementation.

A NOTE OF CAUTION

One thing to remember is that the findings are based on district level and about 27 districts where we could match assessment data with NFHS-5 information. While social and economic factors explain some of the differences in learning, other factors like classroom methods, how the curriculum is taught, teacher training and school conditions might also be involved and need more study.